Why a Robot Is Never Sure: State, Controls and Measurements

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1. Overview

> **Key point:** A robot's sensors and wheels are never exact, so one reading can fit many places and one move can end in several. The robot therefore keeps a probability for every place it could be, instead of a single guess.

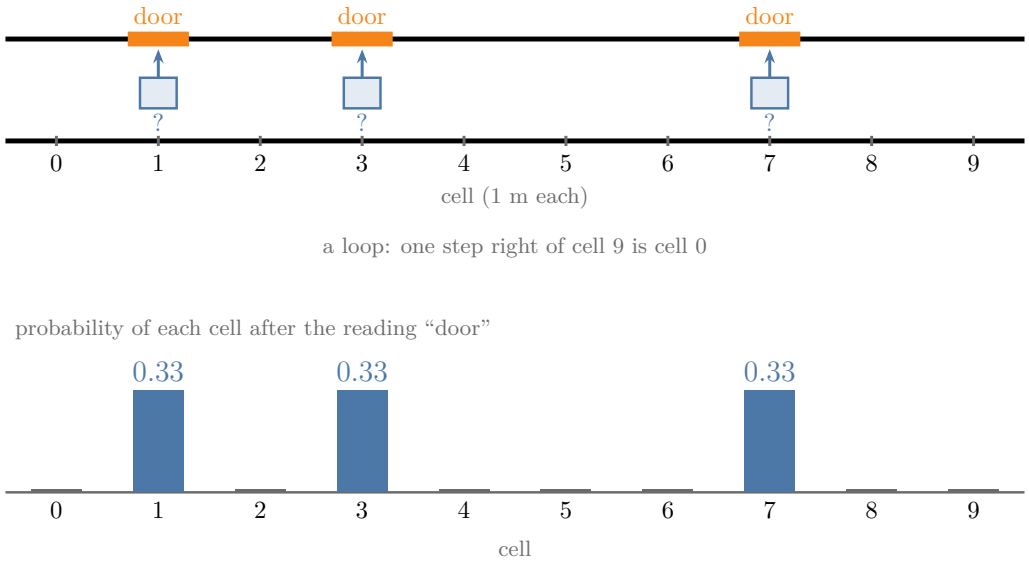


Figure 1: A hallway of 10 cells of 1 m, numbered 0 to 9 and joined into a loop, with doors at cells 1, 3 and 7. The robot's sensor says "door". The robot could be at any of the three doors, so each of those cells gets probability 0.33

So far the kinematic model predicts where wheel commands take a robot: wheels at 12 and 8 rad/s drive it round a circle of radius 0.5 m. That prediction assumes wheels that never slip. Real wheels slip, so a command "move 2" can end 1, 2 or 3 cells on, and a robot that only counts such moves soon does not know where it is. This Note adds the answer: keep a probability for every place. We leave the floor for a hallway of cells, because one number per place keeps the counting small (Section 4.1 returns to the floor).

A delivery robot drives along the hallway of Figure 1, the hallway this whole chapter uses. It has a map: 10 cells of 1 m each, numbered 0 to 9, with doors at cells 1, 3 and 7. The hallway is a loop, so one step to the right of cell 9 is cell 0. The robot has a sensor that says "door" or "wall" for the cell beside it. Its sensor now says "door". Where is it?

There is no single answer. Three cells fit that reading, and the robot has no reason to prefer one of them. A robot that picks one cell and forgets the rest is wrong two times out of three. A robot that keeps all three, each with probability 1/3, still has the true cell among them.

This Note builds that idea step by step:

- why a robot can never be sure: five sources of uncertainty, and how errors pile up when the robot only counts its moves (Section 2);

- why the robot keeps a probability for every place instead of one best guess (Section 3);
- what the robot must keep track of, its state, and when a state is complete (Section 4);
- the two streams of data it receives, controls and measurements, and their order in time (Section 5);
- why moving makes the robot less sure and sensing makes it more sure (Section 6).

2. Why a robot is never sure

> **Key point:** A move of 2 cells can end 1, 2 or 3 cells on, and a reading can be wrong, because every part of a robot adds uncertainty: the world changes, sensors are limited and noisy, wheels slip, models are simplified and computers must answer in time. Errors in counted moves add up, so a robot that only counts its moves gets more and more lost: after 8 moves it is where it thinks only with probability 0.33.

2.1 Why counting moves is not enough: dead reckoning

The simplest way to know where we are is to start at a known place and add up our own moves: "I started at 0 m and moved 2 m eight times, so I am at 16 m." This is **dead reckoning** (G-2311): working out the position from a known start plus the robot's own motion, with no look at the surroundings. Facing north at 1 m/s for 3 s, a robot reckons it is 3 m north of where it started, without checking anything.

The trouble is that the moves themselves are not exact. Our hallway robot is told "move 2", two cells of 1 m. Because its wheels sometimes slip or spin on, the real move is (Labbe ch.2):

Real move	Probability
1 cell (it falls short)	0.1
2 cells	0.8
3 cells (it runs on)	0.1

We count positions in whole cells, so a move that falls well short counts as 1 cell and one that runs on counts as 3. This is the move of the hallway robot throughout the chapter.

Figure 2 plays what happens to a robot that starts surely at 0 m on a long straight track and is told "move 2" eight times without looking around. The bars show the probability of each true position; the red dashed line is where the robot thinks it is.

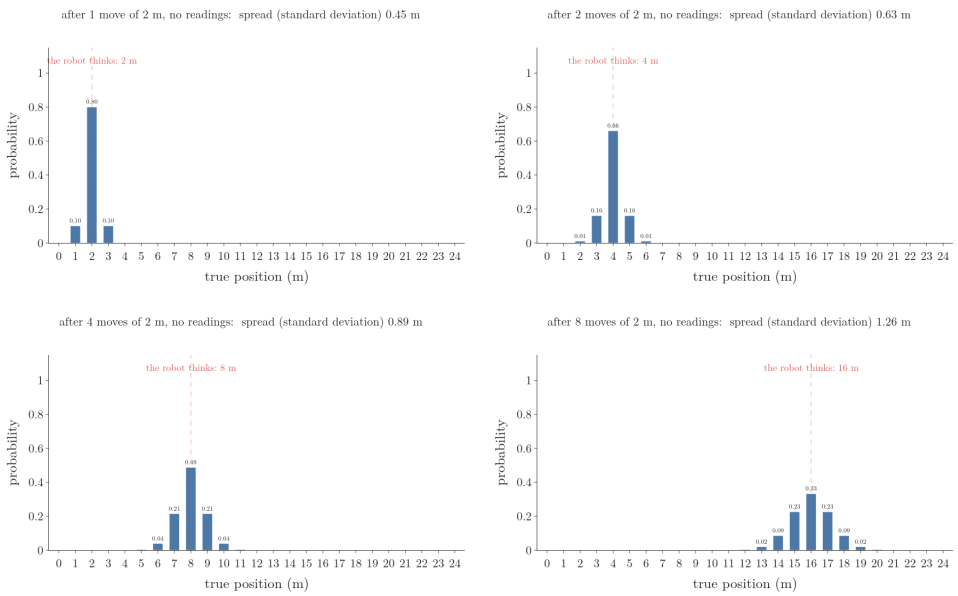


Figure 2: Dead reckoning without readings: after 1, 2, 4 and 8 moves of 2 m the probability of each true position spreads wider; after 8 moves the robot is exactly where it thinks (16 m) with probability 0.33 only

The chance that the robot is exactly where it thinks shrinks with every move:

Moves	Robot thinks	Chance it is exactly there	Spread (standard deviation)
1	2 m	0.80	0.45 m
2	4 m	0.66	0.63 m
4	8 m	0.49	0.89 m
8	16 m	0.33	1.26 m

The spread is the standard deviation (G-1870), the typical distance between the true position and the reckoned one. Why does it grow? One move has variance 0.2 m^2, the average squared miss from the commanded 2 m:

$$0.1 \times (1-2)^2 = 0.1$$

$$0.8 \times (2-2)^2 = 0$$

$$0.1 \times (3-2)^2 = 0.1$$

$$\text{variance} = 0.1 + 0 + 0.1 = 0.2$$

The moves slip independently, and for independent moves the variances add. After 8 moves:

$$\text{variance} = 8 \times 0.2 = 1.6$$

$$\text{spread} = \sqrt{1.6} = 1.26 \text{ m}$$

The spread keeps growing, more slowly than the number of moves but without limit. Dead reckoning is good over short stretches; over longer ones its error grows until something outside the robot corrects it. The notebook `RO-007-why-a-robot-is-never-sure.ipynb` checks the table with 100 000 simulated robots.

2.2 Five sources of uncertainty

Slipping wheels are only one cause. The textbook on probabilistic robotics lists five (Thrun et al. 2005 ch.1, as given in Khamis L5), shown in Figure 3:

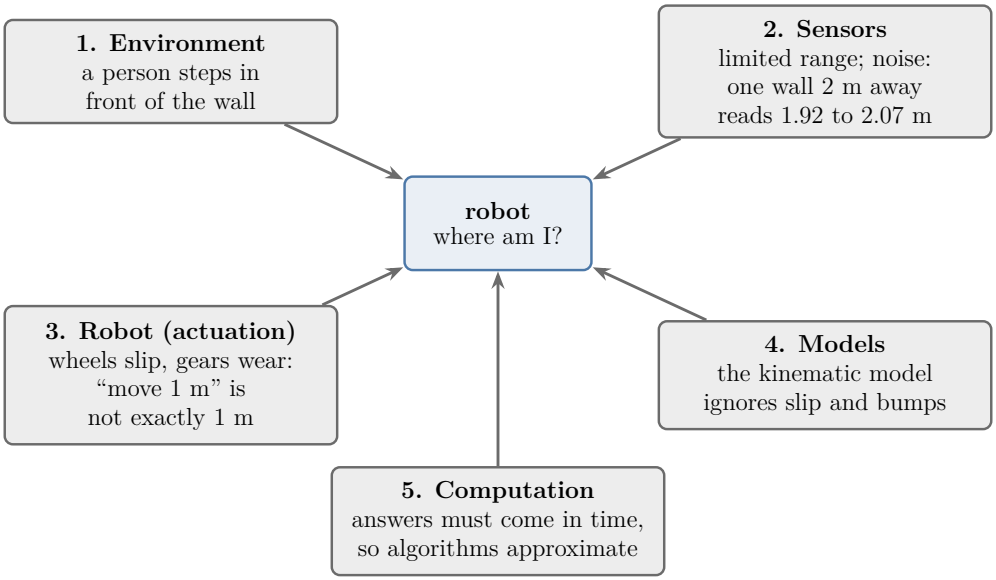


Figure 3: Five sources of uncertainty around a robot: the environment, the sensors, the robot's actuation, the models and the computation, each with an example

1. **The environment.** The world is not fully predictable: people walk past, doors open and close, chairs move. A person who steps between the robot and the wall makes a distance sensor read much less than the true distance to the wall.
2. **The sensors.** A sensor sees only so far and so finely; a camera cannot see through walls. Eight distance sensors around a robot tell it how far the nearest obstacle is in eight directions, which is usually not enough to say where it is in a room. And every reading carries **sensor noise** (G-2312): small random errors, so the same wall measured again from the same spot gives slightly different numbers.

3. **The robot's actuation.** Motors and wheels do not do exactly what they are told. A command to move 1 m gives 95 cm one time and 1.1 m the next; gears slip and wear.
4. **The models.** Every model simplifies. The kinematic model (G-2297), the equations that turn wheel speeds into motion, assumes wheels that never slip.
5. **The computation.** A robot must decide in time, so its algorithms take shortcuts and approximate (Khamis L5).

Figure 4 shows sensor noise and the environment at work on one wall. A distance sensor stands 2.00 m from a wall and takes 200 readings. The 190 readings of the wall itself spread from 1.92 m to 2.07 m, because of sensor noise. The 10 readings taken while a person walked past are between 0.61 m and 1.16 m, nowhere near the wall.

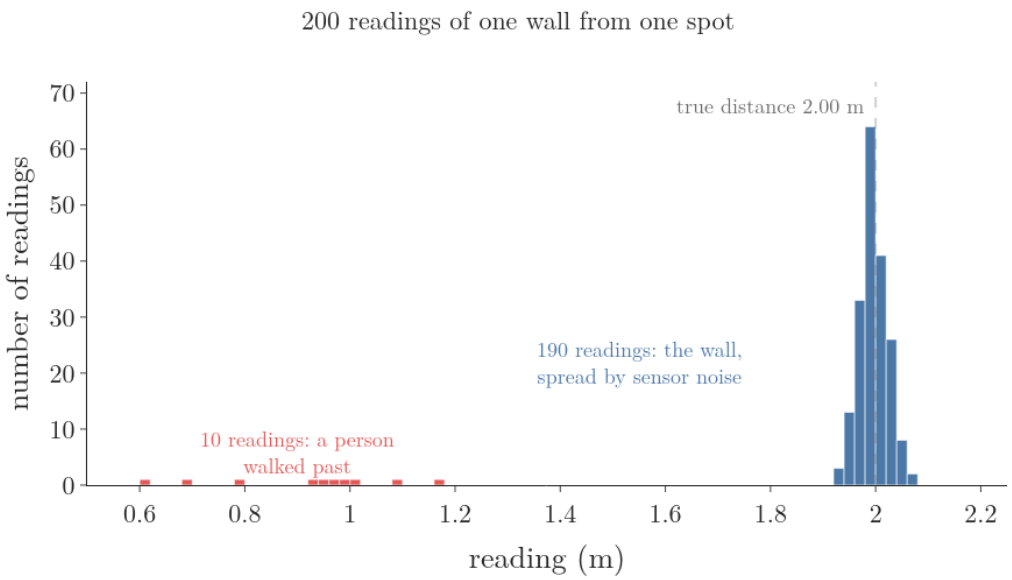


Figure 4: A histogram of 200 readings of one wall from one spot: 190 cluster around 2.00 m between 1.92 and 2.07 m, and 10 readings taken while a person walked past lie between 0.6 and 1.2 m

So no reading and no move can be taken at face value. The rest of this Note is about how a robot can still work out where it is.

3. Keep a distribution, not one best guess

> **Key point:** This section answers the opening question for one reading: "door" fits cells 1, 3 and 7. A robot that keeps one guess throws the other two away and gets lost when it guessed wrong; a robot that keeps a probability for every cell keeps the true one among its candidates.

3.1 Why one reading fits many places

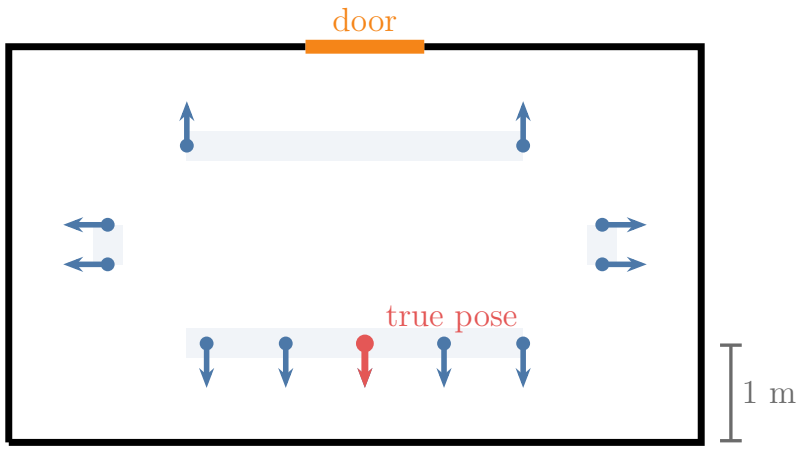
Back to the hallway. For this section we make two simplifications: the moves are exact and the door sensor never errs. At the start the robot knows nothing, so each of the 10 cells has the same probability:

$$P(\text{each cell}) = \frac{1}{10} = 0.1$$

The sensor says "door". Three cells have a door (1, 3 and 7) and nothing favours one of them, so each gets a third:

$$P(1) = P(3) = P(7) = \frac{1}{3} = 0.33$$

The other seven get 0, as Figure 1 shows. The same thing happens in two dimensions. A robot in a room whose laser scan shows "a flat wall 1 m straight ahead" could stand 1 m from any straight stretch of any wall, facing it (Figure 5). The reading rules out spots near a corner or facing the door, where the scan would look different, but it leaves many poses.



scan: a flat wall 1 m straight ahead. Blue: poses that fit it (and every pose in between).

Near a corner the scan would show the corner; facing the door it would show the gap.

Figure 5: A room 7 m by 4 m with a door. The scan shows a flat wall 1 m ahead; every pose 1 m from a straight stretch of wall, facing it, fits the reading, so candidate poses line all four walls

Such a list of every possible value with its probability, adding up to 1, is a probability distribution (G-1571). Ours has three separate peaks, so it is a multimodal (G-1275) distribution: the robot is not in three places at once, it has narrowed its position down to one of three (Labbe ch.2).

3.2 Why a single best guess gets lost

A robot could instead keep only one value: its **best guess**, a point estimate (G-1507) of its position. Figure 6 runs both robots side by side. The true robot is in cell 1 (red square).

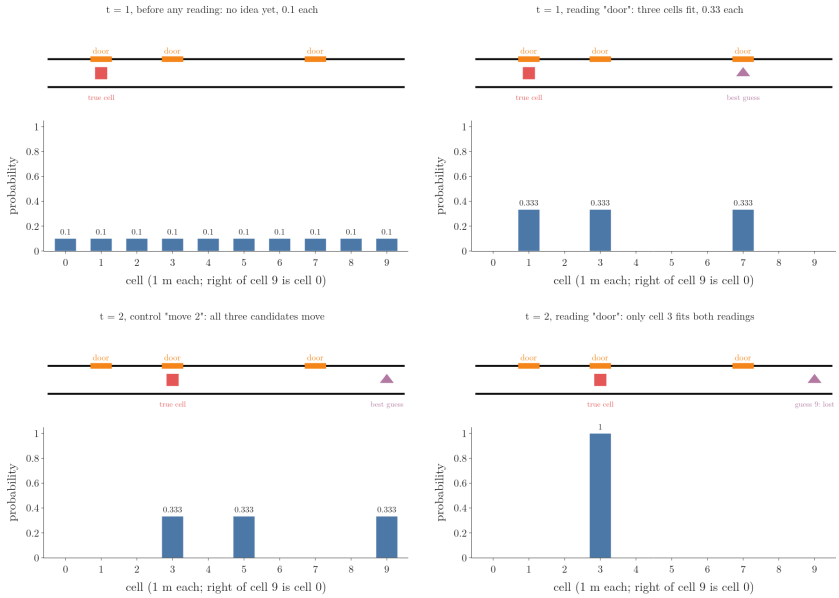


Figure 6: Animation: after the reading "door" the distribution keeps cells 1, 3 and 7 while a best guess picks cell 7; after "move 2" the candidates become cells 3, 5 and 9; after a second "door" only cell 3 fits, while the best guess at cell 9 expected a wall and is lost

Step by step:

1. **Reading "door".** The distribution keeps cells 1, 3 and 7. The best-guess robot must pick one and picks cell 7. With three equal candidates, a guess is right only one time in three.
2. **Control "move 2".** Every candidate moves with the robot: cells 1, 3 and 7 become 3, 5 and 9. The best guess becomes cell 9.
3. **Reading "door" again.** Of cells 3, 5 and 9, only cell 3 has a door. Each other candidate would have read "wall", which does not match. So the distribution is now certain: cell 3, with probability 1.

The best-guess robot expected a wall at cell 9 and read "door". It now knows its guess was wrong, but it threw the other candidates away at step 1, so it has nothing to fall back on: it is lost. The robot that kept the whole distribution found its place, because it remembered every place the readings allowed (Labbe ch.2).

This is the central idea of **probabilistic robotics** (G-2313): the robot computes a probability distribution over what might be the case, instead of a single best guess, so it can handle ambiguous readings and recover from errors (Thrun 2000). The probability distribution a robot keeps over where it might be is its belief, defined later in this chapter.

3.3 Why noisy sensors forbid crossing places out

Section 3.2 crossed out every cell that did not match a reading. That works only for a perfect sensor. A real door sensor errs. The sensor of our hallway robot, used throughout the chapter, behaves like this:

Robot is at	Reads “door”	Reads “wall”
a door cell	0.6	0.4
a wall cell	0.2	0.8

A robot that still crosses out non-matching cells throws away its true cell every time the reading is wrong. In Section 3.2 the true robot stood at a door both times (cells 1 and 3). The chance that both readings say “door” is:

$$0.6 \times 0.6 = 0.36$$

So the chance that at least one says “wall”, and the true cell is crossed out, is:

$$1 - 0.36 = 0.64$$

Nearly two times in three, the robot would delete the cell where it really is. (The two readings err independently (G-934), which is why their probabilities multiply.)

So with a noisy sensor the robot does not cross out; it **weights**. A reading of “door” makes each door cell 3 times as likely as each wall cell, because the sensor reads “door” 3 times as often at a door as at a wall (Labbe ch.2):

$$\frac{0.6}{0.2} = 3$$

Three door cells and seven wall cells make 16 equal shares:

$$3 \times 3 + 7 \times 1 = 16$$

$$P(\text{each door}) = \frac{3}{16} = 0.1875$$

$$P(\text{each wall}) = \frac{1}{16} = 0.0625$$

Check: the ten cells add up to 1.

$$3 \times 0.1875 + 7 \times 0.0625 = 1$$

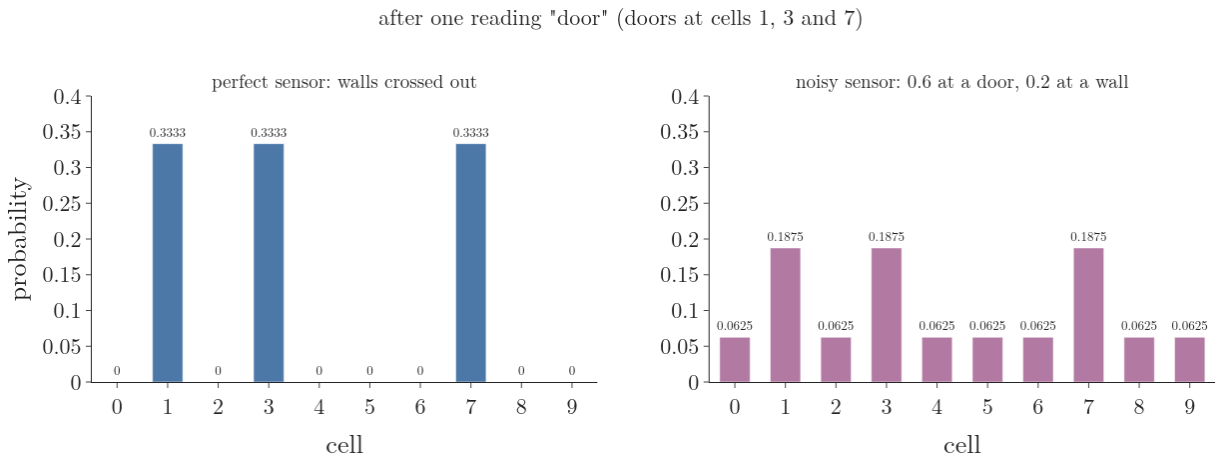


Figure 7: After one reading "door": a perfect sensor gives 0.33 to each door cell and 0 to each wall cell; the noisy sensor gives 0.1875 to each door cell and 0.0625 to each wall cell

Figure 7 compares the two. The wall cells keep a small probability, so one wrong reading lowers the true cell but never deletes it; later readings can lift it again. How a reading turns into these numbers in general is Bayes' theorem, worked out later in this chapter.

4. The state: what the robot keeps track of

> **Key point:** "Where is it?" asks for the robot's state: in the hallway one number, its cell. In general the state is every quantity the robot needs to predict what happens next: its pose, often its speeds, and the things around it that can change. A state is complete when knowing it now makes the past useless for predicting the future.

4.1 State: from one number to many

In the hallway the robot needs one number, its cell. A robot on a floor needs three, its pose (G-2280): position (x, y) and heading θ, such as (2 m, 1 m, 30°).

The collection of quantities the robot keeps track of because they decide what happens next is its **state** (G-2314), written x_t at time t. In the hallway run of Section 3.2 the robot is in cell 3 at time 2:

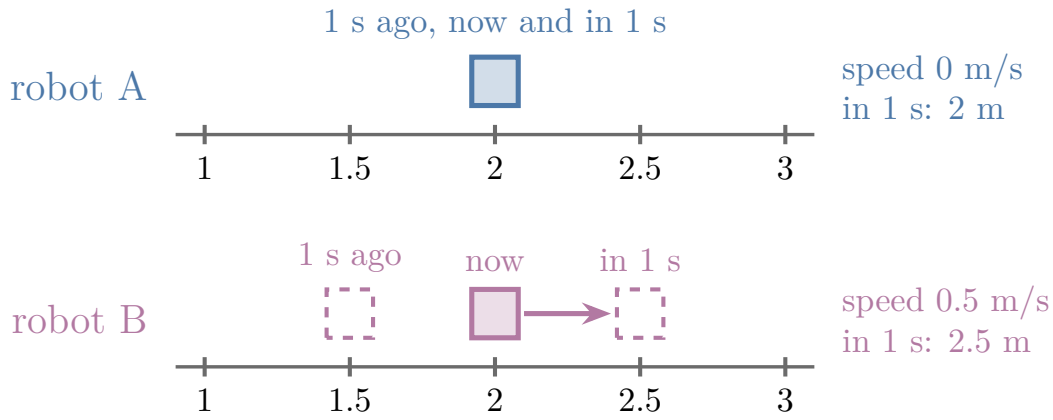
$$x_2 = 3$$

Depending on the robot and the task, the state can also include:

- the **pose** of the robot;
- the **speeds** of the robot, when it does not stop the moment its motors stop (Section 4.2);
- the **positions of things around it**: fixed things such as walls and doors are kept in a map, while things that move, such as people, go into the state, often with their speeds.

4.2 Why the speed can belong to the state

Figure 8 shows two robots on a smooth floor with their motors switched off. Both are at 2 m right now. Robot A has been standing still; robot B is rolling at 0.5 m/s.



both at 2 m now, no command: the speed decides where each robot will be

position alone is not a complete state; (position, speed) is

Figure 8: Two robots both at 2 m now with motors off: robot A stood still and will still be at 2 m in one second; robot B rolls at 0.5 m/s, was at 1.5 m a second ago and will be at 2.5 m

One second later:

Robot	Position now	Speed now	Position in 1 s
A	2 m	0 m/s	2 m
B	2 m	0.5 m/s	2.5 m

Their positions now are the same, but their futures differ. If the state were the position alone, we could not predict where each robot goes, while their positions one second ago (2 m and 1.5 m) would tell us. The past would carry information the state lacks. With the speed added, the state $(2\text{ m}, 0.5\text{ m/s})$ predicts robot B's next position by itself, and the past adds nothing.

4.3 Complete state and the Markov property

That test gives the definition. A state is a **complete state** (G-2315) if knowing it now, the earlier states, controls and measurements tell us nothing more about the future (Thrun et al. 2005 §2.3.1). For robot B:

- position only: not complete, because the position one second ago improves the prediction;
- position and speed: complete, because nothing from the past improves it.

A sequence of states with this property, where the future depends on the past only through the present, has the **Markov property** (G-2316). We want it because it lets the robot forget its history: to predict and to update its belief, it needs only the latest state, not every reading since it was switched on. The Bayes filter is built on it.

In practice no robot's state is truly complete: people move, gears wear, the floor changes. Robots still treat their state as complete, and the methods of this chapter work well when the leftover effects are small (Thrun 2000).

5. Controls and measurements: two streams of data

> **Key point:** The robot answers "where is it?" from two kinds of data. Controls, such as "move 2", change its state; measurements, such as "door", only report on it. A control comes first, then the measurement in the new state.

5.1 Measurements and controls

The data that reaches a robot comes in two kinds (Thrun 2000; Khamis L5):

- A **measurement** (G-2317) z_t is what the sensors report about the world at time t: our door sensor's "door", a laser scan of distances, a camera image. In the hallway at time 1:

$$z_1 = \text{door}$$

- A **control** is information about how the state changed: the command "move 2", or a count of how far the wheels turned. We write it u_t, the same symbol as the control input (G-2298) of a robot model. In the hallway at time 2:

$$u_2 = \text{move } 2$$

We keep them apart because they do opposite things to the robot's knowledge (Section 6). A control causes the state to change; a measurement does not change the state, it only tells the robot something about it.

5.2 The order in time

The robot starts in some state x_0. Then each time step has two parts:

1. a control u_t moves the robot from x_{t-1} to x_t;
2. the robot takes a measurement z_t in the new state x_t.

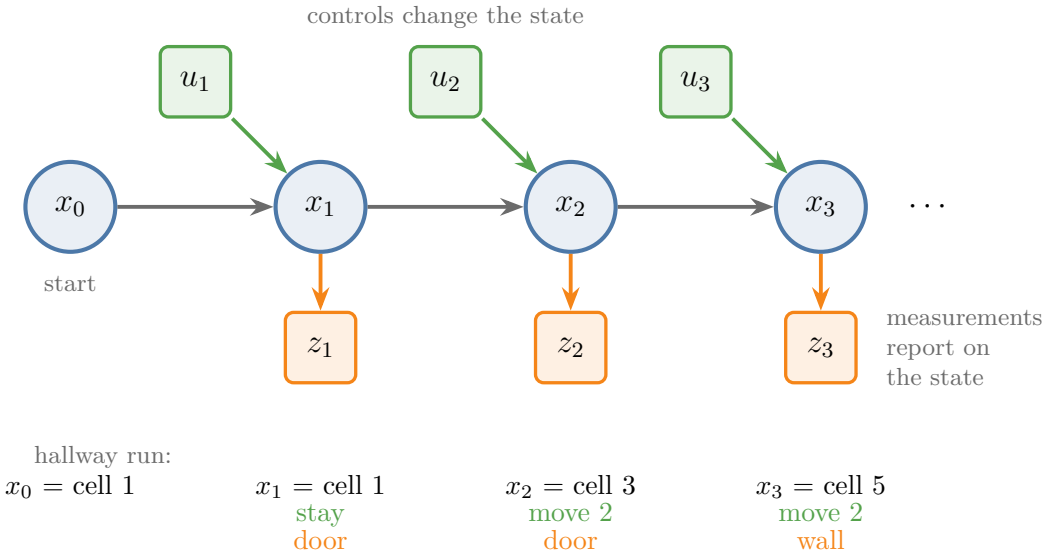


Figure 9: The order of events: from the start state x0, control u1 moves the robot to x1, where measurement z1 is taken; control u2 moves it to x2, where z2 is taken; and so on. Below, the hallway run: from x0 = cell 1, u1 stay gives x1 = cell 1 with z1 door; u2 move 2 gives x2 = cell 3 with z2 door; u3 move 2 gives x3 = cell 5 with z3 wall

Figure 9 draws this chain with the hallway run below it. Controls point into the states (green arrows), because they change them; measurements point out of the states (orange arrows), because each one is produced by the state the robot is in. The robot never sees the states themselves, only the controls it sent and the measurements it received. Working out the states from those two streams is **localization** (G-2318) when the state is the robot's pose in a known map (Thrun 2000).

How likely each next state is after a control, and each reading in a given state, are the two models every method in this chapter needs. The motion model is the first; the measurement model is the second.

6. Moving makes the robot less sure, sensing makes it more sure

> **Key point:** Here the whole opening problem comes back at once: moves that slip and a sensor that errs. Every uncertain move spreads the robot's distribution (from 0 to 0.63 m in two moves); every informative reading narrows it (to 0.42 m). Neither alone is enough: moves alone drift, readings alone are ambiguous, and together they pin the robot down.

Now we drop both simplifications: the moves are the noisy moves of Section 2.1, and the sensor is the noisy sensor of Section 3.3. Take a robot that knows it is in cell 3, as after Section 3.2. It drives two moves of 2 cells without a reading in between, and then reads its sensor. Figure 10 plays the four stages.

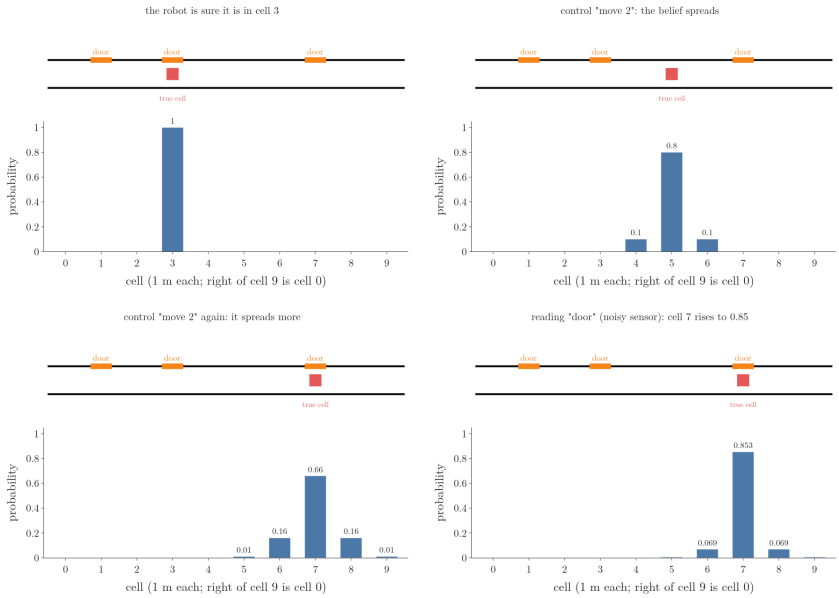


Figure 10: Animation: the robot is sure it is in cell 3; after one noisy move the probabilities are 0.1, 0.8, 0.1 at cells 4, 5, 6; after a second move 0.01, 0.16, 0.66, 0.16, 0.01 at cells 5 to 9; the noisy reading "door" lifts the door cell 7 to 0.853 and lowers cells 6 and 8 to 0.069

Control u_3 spreads the distribution. From cell 3 the robot ends in cell 4, 5 or 6 with the move probabilities of Section 2.1: 0.1, 0.8 and 0.1.

Control u_4 spreads it further. Each way of making two moves multiplies the two move probabilities, because the moves slip independently. Ending in cell 7, four cells on, happens in three ways:

$$2 \text{ then } 2 : 0.8 \times 0.8 = 0.64$$

$$1 \text{ then } 3 : 0.1 \times 0.1 = 0.01$$

$$3 \text{ then } 1 : 0.1 \times 0.1 = 0.01$$

$$P(7) = 0.64 + 0.01 + 0.01 = 0.66$$

The same counting gives the whole distribution:

Cell	5	6	7	8	9
Probability	0.01	0.16	0.66	0.16	0.01

Measurement z_4 = "door" sharpens it. The robot weights each cell as in Section 3.3: it multiplies each probability by the chance of reading "door" there, 0.6 at the door cell 7 and 0.2 at the wall cells:

$$\text{cell } 7 : 0.66 \times 0.6 = 0.396$$

$$\text{cells } 6,\ 8 : 0.16 \times 0.2 = 0.032$$

$$\text{cells } 5,\ 9 : 0.01 \times 0.2 = 0.002$$

What is left no longer adds up to 1:

$$0.396 + 2 \times 0.032 + 2 \times 0.002$$

$$= 0.464$$

Dividing each by that sum gives the probability of each cell given the reading, a conditional probability (G-444):

$$P(7 \mid \text{door}) = \frac{0.396}{0.464} = 0.853$$

$$P(6 \mid \text{door}) = \frac{0.032}{0.464} = 0.069$$

$$P(5 \mid \text{door}) = \frac{0.002}{0.464} = 0.004$$

Cells 8 and 9 get the same as cells 6 and 5. The spread after each stage tells the story in one number (the standard deviation of the distribution):

Stage	Spread
sure in cell 3	0 m
after u_3	0.45 m
after u_4	0.63 m
after z_4	0.42 m

Moves add uncertainty because the robot cannot tell which of the possible outcomes happened; readings remove it because they make places that do not fit less likely, or rule them out (Thrun 2000). That is why a robot needs both streams of data:

- with moves alone it drifts, as in Section 2.1;
- with readings alone it stays ambiguous, as the three doors of Section 3.1 show;
- with both, each reading is checked against where the moves say the robot can be, and the places that fit everything gain probability, as in Section 3.2 and here.

7. Summary

Idea	What it says	Why it matters
Dead reckoning	add up the robot's own moves from a known start	its error grows without limit, so it needs readings to correct it
Five sources of uncertainty	environment, sensors, actuation, models, computation	no reading or move can be taken at face value
Distribution, not best guess	keep a probability for every possible state	the truth stays among the candidates when a reading is ambiguous
State x_t	the quantities that decide what happens next	it is what the robot estimates
Complete state	the past adds nothing once the state is known	the robot can forget its history (Markov property)
Control u_t, measurement z_t	data that changes the state, data that reports on it	controls spread the distribution, measurements narrow it

- "Where is it?" after the reading "door" has no single answer: cells 1, 3 and 7 each get 0.33, because a single guess is right only one time in three. A move and a second "door" then leave only cell 3; with slipping moves and the noisy sensor, two moves and a "door" give cell 7 probability 0.853.
- Dead reckoning drifts: after 8 noisy moves the robot is where it thinks with probability 0.33, because independent errors add up (variances add).
- Uncertainty comes from five places, so a robot must plan for imperfect readings and moves rather than hope for perfect ones.
- A single best guess, once contradicted, has nothing left, so the robot keeps every candidate.
- With a noisy sensor the robot weights places instead of crossing them out, because crossing out would delete the true place whenever a reading is wrong (64 percent of the time over two readings at two doors, with a sensor that sees a door 60 percent of the time).
- The state holds what decides the future: pose, speeds when the robot coasts and moving things around it; it is complete when the past adds nothing, which lets the robot keep only its latest belief.
- A control comes first and changes the state; a measurement follows and reports on it, so each step of a filter is "move, then look".
- Moves spread the distribution (0 to 0.63 m in two moves) and readings narrow it (0.63 to 0.42 m), so the robot needs both.

The robot can now say where it might be as a probability for every cell, and why moves spread it and readings narrow it. Still missing: by how much a move blurs the pose, which the motion model gives.

8. Sources

Built from

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9. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Dead reckoning (G-2311)	Working out a robot's position from a known start by adding up its own moves, without looking at its surroundings; simple, but its error grows with every move, so it needs readings to correct it.

Term	Meaning
Sensor noise (G-2312)	Small random errors in a sensor's readings, so the same quantity measured twice gives slightly different numbers (a wall 2 m away reads 1.92 to 2.07 m); it is one reason a robot cannot take a reading at face value.
Probabilistic robotics (G-2313)	The approach in which a robot keeps a probability distribution over what might be the case (such as where it is) instead of one best guess, so it can handle ambiguous readings and recover from errors.
State x_t (G-2314)	The quantities a robot keeps track of at time t because they decide what happens next, such as its pose, its speeds and moving things around it; it is what localization and filtering estimate.
Complete state (G-2315)	A state that, once known, makes earlier states, controls and measurements useless for predicting the future (position plus speed for a coasting robot, not position alone); it lets the robot keep only its latest estimate.
Markov property (G-2316)	The property of a sequence of states that the future depends on the past only through the present state; it lets a filter forget the history and work from the latest state alone.
Measurement z_t (G-2317)	What a robot's sensors report about the world at time t, such as "door", a laser scan or a camera image; it does not change the state but tells the robot something about it, which narrows its distribution.
Localization (G-2318)	Working out a robot's pose in a known map from the controls it sent and the measurements it received; needed because the pose itself cannot be measured directly.
Standard deviation of a random variable (G-1870)	The square root of a random variable's variance, in the units of X; it says how far values typically fall from the expected value.
Kinematic model (G-2297)	Equations that give how a robot's configuration changes from its speed inputs alone, ignoring masses and forces, such as $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = \omega$; used to predict motion at low speed.
Probability distribution (G-1571)	A list of every possible outcome of a random variable with its probability.

Term	Meaning
Multimodal (G-1275)	Having more than one mode (two modes: bimodal).
Point estimate (G-1507)	A single number computed from sample data as the best guess for an unknown population parameter.
Independent events (G-934)	Events where one happening does not change the probability of the other.
Pose (G-2280)	Where a robot is and which way it faces: on a floor, its position (x, y) and heading θ, such as (2 m, 1 m, 30°); needed because a robot at one spot drives off differently depending on its heading.
Control input u (G-2298)	The numbers we choose to drive a robot, such as $u = (v, \omega)$; the model turns them into motion.
Conditional probability (G-444)	The probability of an event given that another event has happened: $P(A \mid B)$.